

Macroeconomic and Bank-specific Drivers of Non-Performing Loans and the EBA's EU-wide Transparency Exercise: An Empirical Investigation.

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Table of contents

- 1 Introduction
- 2 EBA & SSM
- 3 NPL Drivers & Literature
- 4 Research Question
- 5 Data Description
- 6 Data Visualization
- 7 Empirical Approach
- 8 Results
- 9 Conclusion

Topic

An analysis of the impact of macroeconomic and bank-specific factors on Non-Performing Loans (NPLs) in the EU-banking sector using an extensive, granular dataset on major European banks gathered by the European Banking Authority (EBA) in the context of the EU-wide Transparency Exercise for the period 2014-2019.

Motivation I [11] [12] [14] [7]

- NPLs are a main indicator of bank's asset quality.
- The term NPL usually refers to loans that are over 90 days past-due or unlikely to be repaid in full
- Important effects of financial intermediation on the real economy.
- Understanding the determinants of Non-Performing Loans (NPL) is a very relevant topic for economic policy in Europe. In the last years, non-performing loans have been a major concern for supervisor, policymakers and market participants in the EU.
- Increase importance of understanding NPL determinants in the immediate pre-COVID period and their transmission channels.
- Implications for prudent bank stress testing, EU policy making, recommendations on NPL strategies and design of Early Warning Indicators (EWI), contributing to a more effective microprudential supervision of banks in Europe.

Motivation II - Research on NPL Determinants & Bank-Level Data [21] [9]

- Many authors indicate the need of a more granular analysis of NPL drivers at bank-level.
- Challenges in the NPL literature when using individual bank data:
 - ▶ Data availability is limited due to data protection regulation and banking law.
 - ▶ Private platforms like Dealscan, Datastream, Bankscope or Debtwire suffers from a representativeness bias and data quality issues.
 - ▶ Banking supervisory data is usually provided in a domestically controlled cross-border consolidation basis.
 - ▶ The use of macroeconomic data based on the bank's country of origin does not totally account for cross-border portfolio diversification in the banking sector when analysing its impact on non-performing loans
 - ▶ Divergences concerning the definition of non-performing loans and the method of compilation of the data across countries.
- Example of the potential benefits from using the data of the EBA's EU-wide Transparency Exercise for supervisory and academic purposes with respect to these challenges
- The use of the data of the EBA's EU-wide Transparency Exercise is also quite a novelty in the academic research.

The EBA's EU-wide Transparency Exercise [12] [14]

- The European Banking Authority (EBA) is an independent EU Authority established on 1 January 2011 as part of the European System of Financial Supervision (ESFS) to ensure effective and consistent prudential regulation and supervision across the European banking sector.
- With the EU-wide transparency exercise, the EBA has been disclosing data on an annual basis since 2016, which:
 - ▶ Provides the general public and market participants with updated (quarterly/semesterly) information on the financial conditions of EU banks at the highest level of consolidation
 - ▶ Discloses an extensive, and granular portfolio-by-portfolio supervisory data on major European banks -exposures are broken down by banks, asset classes, default status, ect... representing approximately between 80% and 85% of banking assets in the European Union.
 - ▶ Uses an harmonised definition of NPLs across European countries.
 - ▶ Includes data on original exposures up to the 10 largest countries of counterparty at bank level
 - ▶ Provides information regarding membership status under the Single Supervisory Mechanism (SSM)

The Single Supervisory Mechanism (SSM) [15]

- The Single Supervisory Mechanism (SSM), the system of banking supervision in the euro area, became fully operational on November 4, 2014 as the first pillar of the European banking union.
- The SSM aims to safeguard financial stability and promote European financial integration in cooperation with the established institutional structures.
- Under the SSM, the ECB is responsible for the supervision of banks in the euro area. Non-euro area EU countries may participate in the SSM on a voluntary basis through close cooperation between the competent supervisory authorities.



NPL Drivers

Macroeconomic Factors

- GDP growth
- Unemployment
- Sovereign debt
- Price levels
- Terms of trade
- Legal aspects (rule of law)

Bank-specific Factors

- Cost efficiency
- Bank capitalization
- Bank profitability
- Lending policy
- Economics of information
- Regulation and Supervision
- Geographic diversification

Loan-specific Factors

- Disaggregated loan aspects

NPL Drivers - Literature Review

- Feedback links of financial intermediation and the real economy [6]
- Links between bank-specific characteristics and problem loans:
 - ▶ Bad luck, bad management, skimping hypotheses, too big to fail behavior [5] [26]
 - ▶ Moral hazard hypothesis & risk-taking behavior [10]
 - ▶ Bank capitalization & risk appetite [25] [30]
 - ▶ Strategic aspects of a banks' loan portfolio [18]
 - ▶ Geographic diversification [4]
 - ▶ Disaggregated loan aspects [19]
- Links between Macroeconomic characteristics and problem loans:
 - ▶ Business cycle, terms of trade & price levels [28] [8]
 - ▶ Sovereign debt [31]
 - ▶ Rule of law, Corporate Governance & Supervision [3]
- Usual empirical methods:
 - ▶ Linear panel data methods Dynamic panel data methods: GMM / OLS/VAR
 - ▶ Unbalanced and balanced panel Data estimation techniques
- Type of data:
 - ▶ Aggregate (country) level
 - ▶ Disaggregate (bank) level
- Framework: Geographical (Global/national/regional), Temporal (Quarterly/semesterly/yearly)

Research Question

- **Main question:**

What are the main macroeconomic and bank-specific drivers of Non-Performing Loans in the EU-banking sector?

- **Secondary questions:**

Are there different conclusions given:

- ▶ *Exposure types*
 - Households
 - Non-Financial Corporations
 - Financial Corporations other than credit institutions
- ▶ *NPL levels*
- ▶ *Geographic diversification of exposures*
- ▶ *Supervision under the Single Supervisory Mechanism (SSM)*

Data

- The (unbalanced) sample consists of over 130 banks, at the highest level of consolidation, from 27 countries in the EU and the European Economic Area (EEA) between 2014 and 2019.

Sources:

- *EBA Transparency exercises: Databases published between 2015 and 2020*
<https://eba.europa.eu/risk-analysis-and-data/eu-wide-transparency-exercise>
- *IMF Data*
<https://data.imf.org/?sk=388DFA60-1D26-4ADE-B505-A05A558D9A42&sId=1479329132316>
- *World Bank Data*
<https://datacatalog.worldbank.org/>
- *World Bank Doing Business Report*
<https://www.doingbusiness.org/>

Time period:

2014-2019

Frequency :

Yearly (2014/12,2015/12,2016/12,2017/12,2018/12,2019/12)

Cross-Border Exposure-Weighted Macroeconomic Indicators [27] [16]

- To determine the macroeconomic drivers on NPLs for banks with a cross-border portfolio using data provided on a domestically controlled consolidation basis, the use of cross-border exposure-weighted macroeconomic indicators was required
- The following formula was applied to calculate the Cross-Border Exposure-Weights:

$$y_{it}^* = \sum_{l=0}^{N_{it}} w_{il}^t Y_{lt} \quad \text{with} \quad \sum_{l=0}^{N_{it}} w_{il}^t = 1$$

where w_{il}^t is the cross-border exposure-weight of bank i from country l at time t , Y_{lt} is the macroeconomic variable of country l at time t and y_{it}^* is the cross-border exposure-weighted macroeconomic indicator for bank i at time t . The value of N_{it} varies depending on the number of countries with the largest cross-border exposures for each bank i at time t with $\max(N) = 10$.

Cross-Border Diversification Index (*CBDI*) [29] [20] [4]

- A geographic diversification index of bank exposures is also introduced into the model in order to account for the effects of cross-border diversification on non-performing loans.
- The Cross-Border Diversification Index (*CBDI*) has been calculated as the inverse of the the Hirschmann-Herfindahl Index (*HHI*) for each bank i at time t . Formally,

$$CBDI_{it} = 1 - HHI_{it}$$

with

$$HHI_{it} = \sum_{j=0}^{N_{it}} \left(\frac{X_{ijt}}{X_{it}} \right)^2$$

where X_j measures the original exposure towards country j and N_{it} represents the number of countries with the largest cross-border exposures for each bank i at time t with $\max(N) = 10$. The Hirschmann-Herfindahl indexes are bounded by $1/N_{it} \leq HHI_{it} \leq 1$.

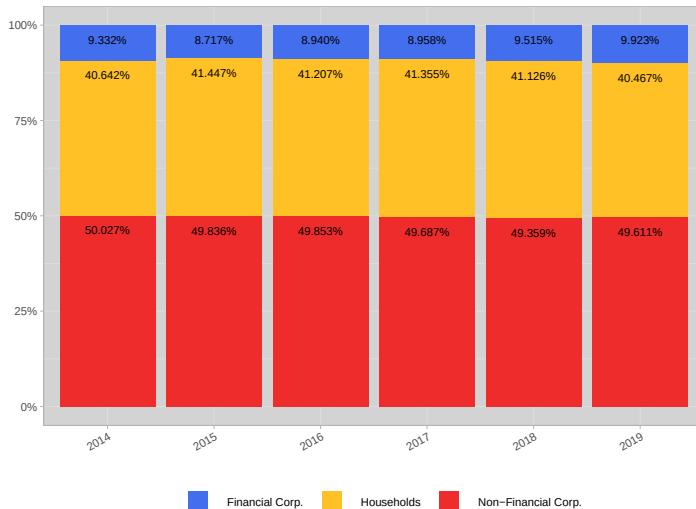
- The Cross-Border Diversification Indexes (*CBDI*) are bounded by $0 \leq CBDI_{it} \leq (N_{it} - 1)/N_{it}$. A Cross-Border Diversification Index (*CBDI*) close to its maximum means that the bank original exposures are very diversified across countries whereas an index close to zero means that the bank original exposures are very concentrated on certain countries

Variables

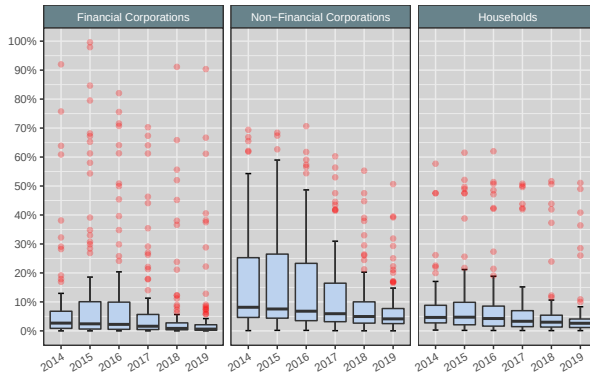
	Variable	Explanation
Bank-Specific	NPL	Non-Performing Loans to total gross loans
	Exposure	Exposure Type
	BankSize_tran	Total Bank Size
	CET1_ratio_tr	Common Equity Tier 1 Ratio
	Cost_to_Income	Cost to Income ratio
	LEVER_tr_ratio	Leverage ratio - using a transitional definition of Tier 1 capital
	NIM_tr	Net Interest Margin ratio
	RORAC	Return on Regulatory (Risk-Adjusted) Capital
Macroeconomic	CBDI_index	Cross-Border Diversification Index (CBDI)
	infl_perc	Inflation, consumer prices (annual %)
	Stgth_legal_index	Strength of legal rights index (0=weak to 12=strong)
	GDP_growth_perc	GDP growth (annual %)
	Reco_rate_dol	Recovery rate (cents on the dollar)
	unempl_rate_imf	Unemployment rate
	curr_acc_bal	Current account balance, percent of GDP
	gross_public_debt	General government gross debt, percent of GDP
	SSM	Single Supervisory Mechanism Dummy

Source: Author's compilation

Total Gross Loans Distribution per Exposure Type (%)



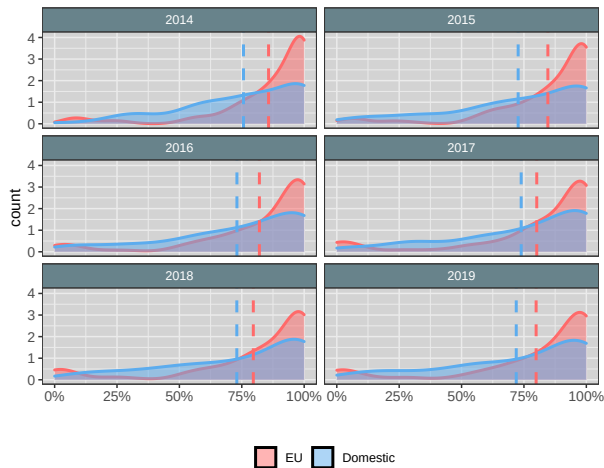
NPL Distribution per Exposure Type



Source: EBA Transparency Exercises, authors' calculations

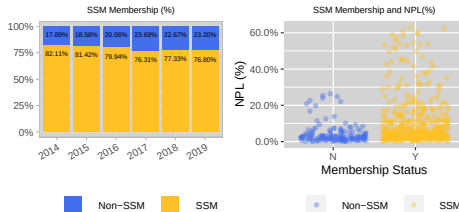
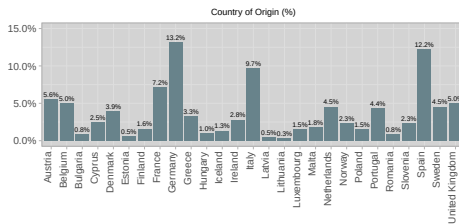
	Financial Corporations	Non-Financial Corporations	Households
Mean	8.339	12.306	6.932
Median	1.532	5.807	3.502
Pctl(25)	0.398	3.177	1.575
Pctl(75)	5.659	16.603	7.119
Standard Deviation	17.432	14.575	10.617
Skewness	2.998	1.874	3.147
Kurtosis	8.839	3.046	9.898

Exposures Ratio Distribution: EU & Domestic



Source: EBA Transparency Exercises, authors' calculations

Single Supervisory Mechanism (%) & Country of Origin (%)



Source: EBA Transparency Exercises, authors' calculations

Linear & GMM Panel Models [2] [1]

- Static Models (Random Effects/Fixed Effects):

$$y_{it} = \alpha + X'_{it}\beta + u_{it} \quad i = 1, \dots, N; \quad t = 1, \dots, T$$

- Dynamic Models-GMM for short panel data (Arellano-Bover Estimators):

$$y_{it} = \delta y_{i,t-1} + X'_{it}\beta + u_{it} \quad i = 1, \dots, N; \quad t = 1, \dots, T$$

- y_{it} represents the non-performing ratio for bank i in year t , α is a scalar, β is $K \times 1$ and X_{it} is the it th observation on K explanatory variables. The K explanatory variables consist of bank-specific and macroeconomic variables. δ is a scalar and $y_{i,t-1}$ is the lagged dependent variable (NPL_ratio).
- u_{it} follows a one-way component model

$$u_{it} = \mu_i + v_{it}$$

where μ_i denotes the unobservable individual bank-specific effects and v_{it} denotes the remainder disturbance.

- $\mu_i \sim \text{IID}(0, \sigma_\mu^2)$ and $v_{it} \sim \text{IID}(0, \sigma_v^2)$ and the error disturbances v_{it} are assumed to be mutually uncorrelated $\mathbb{E}(v_{it} v_{is}) = 0 \forall s \neq t$ and not correlated with the individual effects $\mathbb{E}(\mu_i v_{it}) = 0$.

Pooled Quantil Regression [17] [22]

- Quantile regression consider the relationship between the explanatory variables and the dependent variable using the conditional quantile function $Q_\tau(y|x)$.
- Quantil regression shows more robutness in samples with many outliers than OLS and it is semiparametric, i.e. it does not make assumptions about the parametric distribution of the errors.
- Quantile regression methods also allows the analysis of the impact of a set of explanatory variables on the entire conditional distribution of the dependent variable.
- The quantile $\tau \in (0,1)$ determines the point of y in which the data is split into proportions under τ and above $1 - \tau$. Quantile regression seeks to minimize the errors ε_i that gives asymmetric penalties $(1 - \tau)|\varepsilon_i|$ for overprediction and $\tau|\varepsilon_i|$ for underprediction.
- The quantile regression estimator in quantile τ is the one that minimizes the following objective function:

$$Q(\beta_\tau) = \sum_{y_i \geq x_i' \beta}^n \tau |y_i - x_i' \beta_\tau| + \sum_{y_i < x_i' \beta}^n (1 - \tau) |y_i - x_i' \beta_\tau|$$

- The estimator have been showed to be asymptotically normal and bootstrap standard errors are often used instead of traditional standard errors.

Fixed Effects Panel Quantil Regression (Penalized FE Method) [23] [24]

- The penalized least squares interpretation offers the possibility of perform quantile regression models accounting for the existence of individual effects in panel data.
- In the Penalized FE Method, the following penalized estimator $(\hat{\alpha}(\lambda), \hat{\beta}(\lambda, \tau))$ is considered

$$\min_{(\alpha, \beta)} \frac{1}{nT} \sum_{k=1}^q \sum_{i=1}^n \sum_{t=1}^T \omega_k \rho_{\tau}(y_{it} - \alpha_i - X'_{it} \beta(\tau_k)) + \lambda \sum_{i=1}^n |\alpha_i|$$

where $\omega_1, \dots, \omega_q$ are nonnegative weights and $\lambda \geq 0$ is the penalty level.

- The idea in this model is to estimate several τ simultaneously and restrict the individual effect coefficients α_i to be the same for all quantiles and which values shrinks them toward zero as the penalty level λ increases. Therefore, the parameter λ allows a calibration of the impact of the penalty.
- In general, when $\lambda \rightarrow 0$ the model is similar to a (weighted) dummy-variable quantile regression and when $\lambda \rightarrow \infty$ the model is more like a (weighted) cross-section quantile regression, i.e, the lower the value of λ , the larger the control for fixed effects in the model.
- The penalty allows a "control" over the many fixed effects coefficients by not letting them reach unrealistic large numbers.

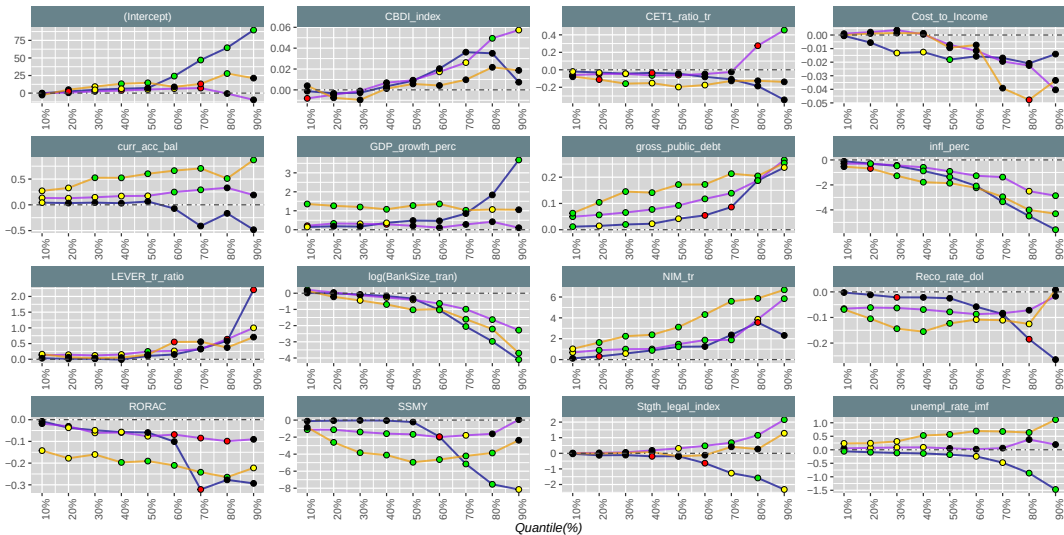
Linear & Dynamic Panel Regressions

	panel linear						panel GMM		
	Households		Non-Financial Corp.		Financial Corp.		Households	Non-Financial Corp.	Financial Corp.
	RE	FE	RE	FE	RE	FE	SYS	SYS	SYS
lag(NPL.ratio)							1.040*** (0.090)	0.791*** (0.081)	0.542*** (0.153)
CET1.ratio.tr	-0.158*** (0.053)	-0.201*** (0.061)	-0.092 (0.103)	0.080 (0.092)	-0.159 (0.142)	-0.020 (0.314)	0.081 (0.071)	0.001 (0.115)	0.024 (0.114)
LEVER_tr.ratio	0.422** (0.192)	0.153 (0.246)	-0.018 (0.272)	-0.592 (0.418)	0.649 (0.626)	0.260 (0.883)	-0.004 (0.098)	-0.071 (0.160)	0.150 (0.368)
RORAC	-0.036** (0.017)	-0.032** (0.016)	-0.027 (0.043)	-0.001 (0.025)	-0.117** (0.049)	-0.042 (0.036)	-0.073** (0.032)	-0.114* (0.064)	-0.270* (0.158)
log(BankSize.tran)	-2.060*** (0.462)	-4.460*** (1.170)	-2.710*** (0.620)	0.514 (1.500)	-1.760** (0.794)	9.850*** (3.530)	0.161 (0.129)	-0.163 (0.223)	-0.231 (0.560)
Cost.to.Income	-0.004 (0.006)	-0.005 (0.005)	-0.013 (0.012)	-0.013 (0.009)	0.019 (0.022)	0.025 (0.029)	-0.004 (0.005)	-0.012 (0.013)	-0.012** (0.006)
NIM_tr	0.758 (0.535)	0.301 (0.483)	1.340 (0.981)	0.492 (1.070)	0.681 (2.160)	0.589 (2.600)	-0.184 (0.733)	1.050 (0.924)	0.296 (1.190)
Stgth_legal.index	0.766*** (0.285)	0.262 (0.220)	1.070** (0.468)	2.230*** (0.419)	0.020 (0.751)	0.820 (0.738)	0.083 (0.163)	0.094 (0.246)	-0.517 (0.542)
Reco.rate.dol	-0.052** (0.022)	-0.023 (0.020)	-0.130** (0.060)	-0.132** (0.067)	-0.175 (0.122)	-0.167 (0.138)	-0.033* (0.017)	-0.090*** (0.032)	-0.120* (0.071)
infl.perc	-0.213 (0.174)	-0.150 (0.139)	-0.637** (0.325)	0.174 (0.279)	-1.190 (0.895)	-0.262 (0.902)	0.209 (0.508)	-0.927 (0.610)	-1.050 (1.100)
curr_acc_bal	0.102 (0.109)	0.186** (0.093)	0.451*** (0.166)	0.512*** (0.200)	0.230 (0.356)	1.060*** (0.384)	0.196*** (0.056)	-0.014 (0.109)	-0.226 (0.244)
GDP_growth.perc	0.244 (0.149)	0.176* (0.102)	0.312 (0.252)	0.043 (0.286)	0.631 (0.725)	-0.149 (0.327)	-0.200 (0.227)	-0.760** (0.330)	0.325 (0.961)
unempl.rate_jmf	0.411*** (0.082)	0.413*** (0.071)	1.660*** (0.183)	2.220*** (0.184)	0.362 (0.221)	1.070*** (0.266)	0.121** (0.057)	-0.103 (0.114)	-0.264 (0.188)
gross_public_debt	0.153*** (0.025)	0.075*** (0.025)	0.195*** (0.039)	0.191*** (0.061)	0.151*** (0.057)	0.063 (0.093)	0.002 (0.015)	-0.001 (0.028)	0.012 (0.045)
SSMY	-1.970 (1.300)		-5.350*** (1.600)		-2.490 (2.790)				
CBDI_index	0.050*** (0.019)	0.031 (0.019)	0.057** (0.026)	0.119*** (0.039)	0.038 (0.040)	0.039 (0.063)	0.014* (0.008)	-0.005 (0.010)	-0.014 (0.023)
Constant	12.800*** (4.940)		19.500** (8.530)		22.700 (16.300)				
Hausman Test	17.51		1202.57		4.69				
Prob (Hausman Test)	0.2298		0		0.9896				
Sargan Test							39.87	43.28	42.09
Prob (Sargan Test)							0.0527	0.0245	0.0322
Arellano-Bond:AR(1)							-1.94	-0.97	-6.36
Prob (AR(1))							0.0525	0.334	0
Arellano-Bond:AR(2)							0.58	-0.93	0.67
Prob (AR(2))							0.5627	0.3501	0.5019
Observations	619	619	628	628	580	580			
R ²	0.376	0.338	0.532	0.575	0.150	0.124			
Adjusted R ²	0.361	0.155	0.521	0.456	0.127	-0.134			
F Statistic	359.000***	17.700***	687.000***	47.300***	95.400***	4.540***			

Note: HC-robust standard errors are used
RE: Random Effects, FE: Fixed Effects, SYS: System-GMM

*p<0.1; **p<0.05; ***p<0.01

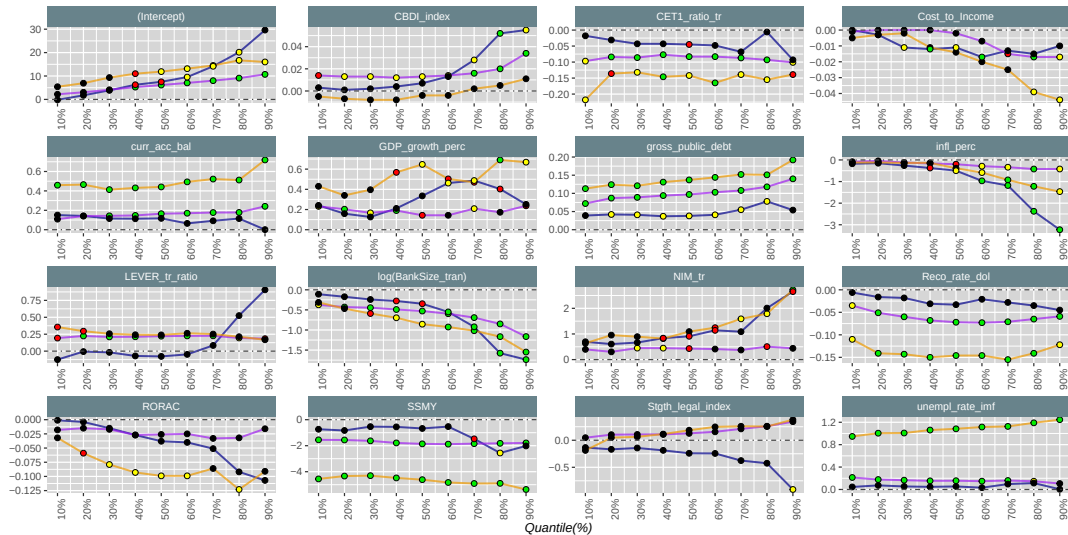
Pooled Quantile Regression



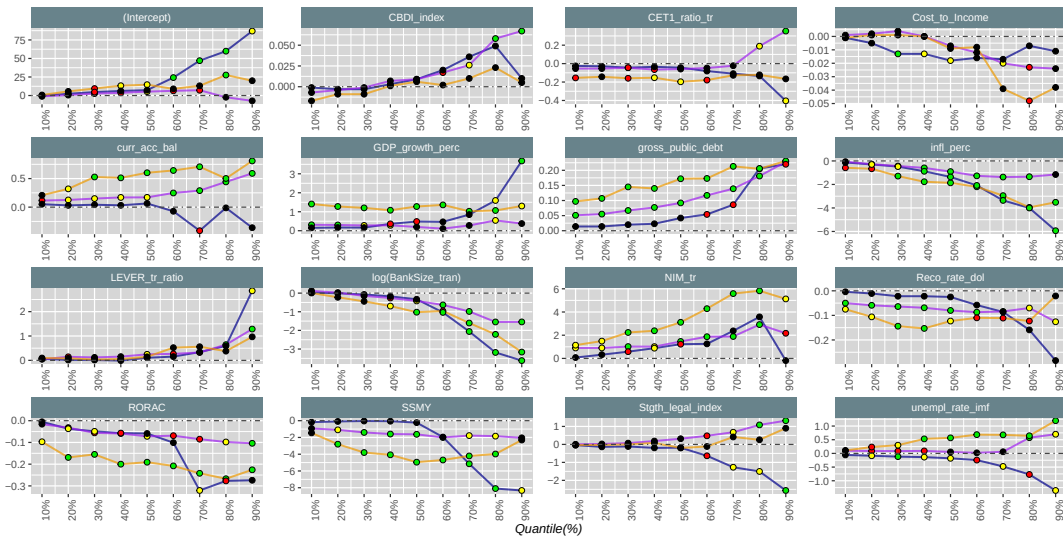
Exposure: — Financial Corp. — Households — Non-Financial Corp.

P-values: ● <0.01 ● <0.05 ● <0.1 ● >=0.1

Penalized Fixed Effects Panel Quantile Regression ($\lambda = 1$)



Penalized Fixed Effects Panel Quantile Regression ($\lambda = 5$)



Exposure

Financial Corp.

Households

Non-Financial Corp.

P-values:

 <0.01 <0.05 <0.1 ≥ 0.1

Results I

- In general, macroeconomic drivers tend to show slightly more explanatory power than bank-specific variables
- **Bank-specific Drivers**
 - ▶ Bank profitability → Returns on Risk adjusted Capital (-)
 - ▶ Cost efficiency → Cost to Income (?)
 - ▶ Bank capitalization → CET1 ratio(-), leverage ratio (+/-)
 - ▶ Lending policy → Net Interest Margin (+/?)
- **Macroeconomic drivers**
 - ▶ Business cycle → unemployment rates(+), & GDP growth (+/-),
 - ▶ Terms of trade → current account balance (+/?)
 - ▶ Sovereign Debt → Gross Public debt (+)
 - ▶ Rule of law:creditor-protection → Recovery rates (-)
 - ▶ Rule of law:creditor & debtor-protection → strength legal system (+/?)
 - ▶ Corporate Governance & Supervision → SSMY(-)
 - ▶ Price levels → inflation (+)

Results II

- **Disaggregated loan aspects**

- ▶ Exposure types (households & non-financial corp.) showed similar effects in term of impact direction (+/-) but differences in intensity and explanatory power, with financial corp. showing the greatest impact direction deviations and the least explanatory power.
- ▶ Greatest impact on non-financial Corporations followed by households & financial corporations
- ▶ Similar marginal changes for each portfolio type
- ▶ In general, the marginal impact increases along the conditional NPL-distribution for all portfolio types, showing that the intensity of the impact increases for banks with higher non-performing ratios

- Positive impact of lagged NPL ratios

- The effects of bank-specific and macroeconomic drivers increase **along the conditional NPL-distribution**, i.e., stronger impacts as the NPL levels in portfolio increase

- **Regulation and Supervision under the Single Supervisory Mechanism** is significant and negatively correlated with NPL with similar effects along the conditional NPL distribution

- The effect of **geographic diversification** are unclear

Conclusion

- Contribute with a more disaggregated analysis of determinants of NPL based on bank-portfolio composition, using novel supervisory data provided by EBA showing:
 - ▶ Importance of bank-specific & regulatory ratios on explaining NPLs in terms of bank-profitability, bank-capitalization, portfolio strategy and risk appetite
 - ▶ Importance of macroeconomic indicators taking geographic composition in portfolio into consideration
 - ▶ A first estimation of the impact of geographic diversification on NPLs
 - ▶ Relevance of regulatory supervision & legal aspects on explaining NPLs
 - ▶ Importance of disaggregated loan-portfolios in NPLs research
 - ▶ Importance of analysing NPL determinants conditional on NPL levels
- Example of potential uses of the data from the EBA transparency exercises in academic research
- Very few papers using quantil regression methodology for panel data in NPL literature
- Paper also provides interesting insights for policy making, the design of Early Warning Indicators (EWI), the preparation of recommendations on NPL strategies and bank-stress testing methodology.

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